Four-Center Multi-Reader Validation + Tier-3 Algorithmic Utility Evaluation
Core Tier-2 Feature Fidelity Assessment for Synthetic Histopathology Images

1. Scientific Rationale

Synthetic histopathology images are increasingly used for augmentation, privacy-preserving sharing, and AI training. However, perceptual realism (Tier-1) does not guarantee preservation of diagnostically relevant features (Tier-2). This document specifies (i) a four-center, blinded reader study to validate a concise Tier-2 protocol and (ii) a Tier-3 algorithmic evaluation assessing downstream utility using multiple foundation-model feature extractors.

2. Research Questions and Hypotheses

2.1 Primary Research Question (Multi-center reader study)

Is the Core Tier-2 scoring protocol reproducible across institutions and does it reliably distinguish real from synthetic patches based on feature fidelity?

2.2 Primary Hypothesis

H1: Inter-rater reliability of the Core Tier-2 total score is high across centers.

$$\text{Target: ICC} \geq 0.75$$

2.3 Secondary Hypotheses

- H2: Center effects are small relative to between-image variability.
- H3: Synthetic patches have lower Tier-2 scores than real patches.
- H4: Key failure modes (e.g., stroma/ECM, TIL plausibility, synthetic artifacts) are consistently detected across centers.

2.4 Tier-3 Research Question (algorithmic utility)

Does training with synthetic augmentation improve downstream performance and generalization to external centers across multiple foundation-model feature pipelines?

2.5 Tier-3 Hypotheses

- H5: Synthetic augmentation improves out-of-center performance (Centers B–D) relative to real-only training.
- H6: Improvements are consistent across multiple foundation-model feature extractors (robustness to representation choice).

3. Study Overview

- **Tier-2 (Human):** Four-center reader study (4 centers × 2 readers = 8 readers), blinded, using a centrally curated patch set ($N = 180$).
- **Tier-3 (Algorithmic):** Single-center training (Center A) with external testing on Centers B–D, repeated across 4 foundation-model feature extractors.

4. Tier-2 Multi-Center Reader Study Protocol (Option A, 4×2)

4.1 Design

- Prospective, blinded, multi-center reader study
- 4 centers: A, B, C, D
- 2 readers per center (target: 1 senior pathologist + 1 trainee/fellow)
- Two sessions of 90 patches to reduce fatigue

4.2 Dataset Construction

$$N = 180, \quad N_{\text{real}} = 90, \quad N_{\text{synthetic}} = 90.$$

Tissue categories (4)

1. Tumor-dominant
2. Stroma-dominant
3. TIL/Inflammation-dominant
4. Mixed / Transition

Real patches (90): by tissue category

Tissue Type	Count
Tumor-dominant	24
Stroma-dominant	24
TIL/Inflammation-dominant	24
Mixed / Transition	18

Synthetic patches (90): tissue category × quality band Synthetic patches are stratified by **quality tertiles** (High/Medium/Low) using a pre-specified automated proxy (e.g., embedding similarity or internal quality score). Sampling is random within strata to prevent confounding.

Per quality band ($N = 30$):

Tissue Type	Count per band
Tumor-dominant	8
Stroma-dominant	8
TIL/Inflammation-dominant	8
Mixed / Transition	6

Optional repeats (recommended) Include $N_{\text{repeat}} = 18$ repeats (10% of N) for intra-rater consistency:

$$9 \text{ real} + 9 \text{ synthetic},$$

balanced across tissue categories and (for synthetic) across quality bands. Repeats are inserted at random positions.

4.3 The Concrete Question(s) Asked to Raters

Session-level instruction (shown once) **Instruction:** You will evaluate histopathology patches. Please assess **feature fidelity** and **histologic consistency** at the level of diagnostically relevant structures. **Do not** attempt to decide whether a patch is real or synthetic.

Per-patch umbrella question **Umbrella question:** *"Are the diagnostically relevant features in this patch consistent with real H&E histology at this magnification, without non-physiologic patterns?"*

Per-patch structured questions (Core Tier-2, 6 items) Each item is scored on a 3-point ordinal scale:

0 = inconsistent / clearly non-physiologic, 1 = borderline / minor inconsistencies, 2 = consistent with real

1. **F1 Nuclear detail & cytology:** Are nuclear size/shape, chromatin texture, and nucleoli patterns consistent with real histology (no impossible nuclear patterns)?
2. **F2 Tissue architecture:** Is the tissue organization (tumor/stroma boundaries, gland/nest structure, transitions) coherent (no broken topology or impossible transitions)?
3. **F3 Stroma / ECM realism:** Is stromal/ECM texture consistent (no "plastic" appearance, melted background, or repeated fibrous motifs)?
4. **F4 Inflammation / TIL plausibility:** Are inflammatory cells/TIL morphology and spatial distribution consistent (no "stamped" lymphocytes or unnatural clustering)?
5. **F5 Synthetic artifacts / hallucination markers:** Are there visible synthetic artifacts (repetition, tiling seams, edge artifacts, non-biologic textures)? (0=clear artifacts; 1=possible/subtle; 2=none visible)
6. **F6 Global clinical usability:** Would you trust this patch enough to use it as evidence in diagnostic reasoning? (0=no; 1=limited/uncertain; 2=yes)

Optional single "main failure reason" item (recommended) If any item was scored 0, select the **main reason**:

- Nuclear/cytology inconsistency (F1)
- Architecture incoherence (F2)
- Stroma/ECM artifact (F3)
- TIL/inflammation artifact (F4)
- Obvious synthetic artifacts/repetition (F5)
- Other (free text)

4.4 Total Score and Predefined Pass/Fail

$$S_{\text{total}} = \sum_{i=1}^{6} F_i, \quad S_{\text{total}} \in \{0, \ldots, 12\}.$$

Predefined Pass/Fail rule (agreement analysis)

- **Pass** if $S_{\text{total}} \geq 9$ *and* $F5 \geq 1$
- **Fail** otherwise

4.5 Calibration, Blinding, and Procedure

- Calibration: 15 annotated calibration patches (10–15 minutes).
- Blinding: readers blinded to real/synthetic label, quality band, and source site.
- Procedure: two blocks of 90 patches with recommended break.
- Data capture: reader_id, center_id, patch_id, F1–F6 (0/1/2), optional uncertainty flag, optional timestamps.

4.6 Tier-2 Endpoints

Primary

- ICC for S_{total} (two-way random effects, absolute agreement) with 95% CI.
- Fleiss' κ for Pass/Fail with 95% CI.

Secondary

- Feature-wise agreement (ICC or κ per feature).
- Center effects via mixed-effects modeling.
- Failure-mode consistency across centers (distribution of "main failure reason").
- Optional intra-rater reliability on repeated items.

4.7 Tier-2 Statistical Model (center effects)

$$S_{\text{total}} \sim \text{center} + (1|\text{image}) + (1|\text{center:reader})$$

Report center contrasts (effect size, CI). Conclude robust multi-center reproducibility if center effects are small relative to between-image variability.

5. Tier-3 Algorithmic Evaluation Plan (Downstream Utility)

5.1 Design Principles

- Training conducted at **one center** (Center A) to create a strong domain-shift stress test.
- External evaluation on **held-out centers** (Centers B–D) with no further tuning.
- Repeated across **four foundation-model feature extractors** to test robustness to representation choice.
- Strict leakage controls: synthetic generation uses **Center A training split only**.

5.2 Task Definition

Specify one primary downstream task (choose based on available labels):

- Slide-level classification (preferred): e.g., tumor vs non-tumor; subtype; grade component proxy.
- Patch-level classification (if only patch labels exist).

All splits must be at **patient level** (no patient overlap between train/val/test).

5.3 Data Splits

- Center A: train/val split (e.g., 80/20) for model development and early stopping.
- Centers B–D: external test sets (100% held-out), used only once for final reporting.
- No stain/scanner adaptation on B–D beyond fixed preprocessing determined on Center A.

5.4 Training Conditions (per foundation model)

For each representation pipeline, train identical downstream classifiers under:

1. **Real-only:** train on Center A real data only.
2. **Real + Synthetic augmentation:** add synthetic samples generated from Center A train split (stratified by tissue type and quality bands).
3. **(Optional) Synthetic-only:** stress test; train only on synthetic, evaluate on real.

5.5 Four Foundation-Model Feature Pipelines

Use four distinct, well-known pathology (or general) foundation models as fixed feature extractors. For each model $m \in \{1, 2, 3, 4\}$:

- Extract tile/patch embeddings for all WSIs/patches using m.
- Aggregate to slide-level (if needed) using a standard method (e.g., mean pooling, attention pooling, or MIL head).
- Train a lightweight classifier (e.g., logistic regression / linear layer / shallow MLP) on Center A train split.

Implementation note: To avoid reviewer debates, keep the downstream head identical across models wherever possible; the goal is to test robustness to representation choice, not to optimize architecture.

5.6 Evaluation Metrics

Primary (report per center B, C, D and pooled):

- AUC (or balanced accuracy if binary not appropriate)
- Sensitivity at fixed specificity (clinically interpretable operating point)
- Calibration: Brier score or ECE (Expected Calibration Error)
- 95% CIs via patient-level bootstrap

Secondary:

- Performance vs training-set size (learning curve): $n \in \{10, 20, 50, 100\}$ (if feasible).
- Robustness across tissue strata (tumor/stroma/TIL/mixed).

5.7 Leakage and Memorization Controls (must-have with synthetic data)

- Synthetic data must be generated from Center A **train** split only.
- Nearest-neighbor audit: compute embedding similarity between each synthetic sample and all real samples in:
  - Center A test split (if any),
  - Centers B–D test sets.

  Flag and remove near-duplicates above a pre-specified similarity threshold.
- Report duplicate rate and confirm no overlap by patient/slide identifiers.

5.8 Tier-3 Primary Endpoint (decision criterion)

Tier-3 "utility" is supported if, for at least 3/4 foundation models, the Real+Synthetic condition improves external-center performance:

$$\Delta\text{AUC}_{\text{ext}} > 0$$

with consistent direction across B–D (or pooled), and without worsening calibration beyond a pre-specified tolerance.

6. Risk Mitigation

- Improve reader consistency: calibration + anchored wording ("consistent with real histology").
- Reduce fatigue: two-session design.
- Prevent cherry-picking: stratified random sampling rule fixed *a priori*.
- Prevent post-hoc tuning: thresholds fixed before analysis.
- Prevent leakage: strict train-only synthesis + nearest-neighbor audits.

7. Ethics and Data Protection

Only de-identified patches/embeddings are used. No patient-identifying information is collected. Local exemption/waiver determination may be sought if required for reader participation.

8. Timeline

- Week 1: finalize patch set, strata labels, survey/tool setup; define Tier-3 task and splits.
- Week 2: reader pilot (internal); implement embedding extraction pipelines for 4 foundation models.
- Weeks 3–4: multi-center reader scoring; run Tier-3 training (Center A).
- Week 5: analysis (Tier-2 ICC/κ; Tier-3 metrics + CIs; leakage audit).
- Week 6: integrate results into manuscript + supplements (scoring guide, calibration set description).

9. Planned Outputs

- Table: dataset composition (real/synthetic, tissue strata, quality bands).
- Table: Tier-2 reliability (ICC, κ) overall and per feature.
- Figure: Tier-2 score distributions (real vs synthetic; by synthetic quality band).
- Figure: center effects (estimated marginal means and contrasts).
- Table/Figure: Tier-3 performance across centers and across 4 foundation models (real-only vs real+synthetic).
- Appendix: full reader instruction sheet + calibration examples; leakage audit description.